

Memory and Processors

1. Memory

Memory is used to store instructions so that the processor can quickly fetch, decode, and execute them. It is accessed much faster than secondary storage.

- **RAM (Random Access Memory):**
 - **Function:** Stores instructions and programs currently in use.
 - **Volatility:** It is **volatile**, meaning all stored data is lost when the power is turned off.
 - **Virtual Memory:** When RAM is full, the computer uses a section of secondary storage as "virtual memory." This slows down system performance significantly because the processor must fetch data from the slower secondary storage instead of the RAM.
 - **Capacity:** Increasing RAM allows more programs to run simultaneously, reducing the need for virtual memory.
- **ROM (Read Only Memory):**
 - **Function:** Stores information required to boot the system or data for single-use devices (e.g., calculators).
 - **Volatility:** It is **non-volatile**; data is retained even when power is lost.
 - **Variations:**
 - **PROM (Programmable ROM):** Data can be written only once.
 - **EPROM (Erasable PROM):** Data can be erased and rewritten using ultraviolet (UV) light.
 - **EEPROM (Electrically Erasable PROM):** Data can be erased and rewritten by applying a high voltage to a specific pin.
- **Flash Memory:**
 - A type of EEPROM that is portable and has no moving parts, resulting in low power consumption. It is commonly used in devices like laptops, USB drives, and SD cards.

2. Processors

The processor is the "brain" of the computer, responsible for executing instructions.

- **The Processor Cycle:** The cycle by which a processor **fetches** from memory, **decodes**, and **executes** instructions.
- **Cores:** A processor may contain multiple CPUs, each referred to as a "core" (e.g., a quad-core processor has four CPUs).
- **Speed:** Processor speed is measured in **Hertz (Hz)**, which represents the number of clock cycles (or instructions carried out) per second.

1. Definitions: Explain the difference between volatile and non-volatile memory. Give one example of each.
2. Performance Analysis: A student complains that their computer slows down when they have many applications open. Explain why this happens and describe the role of virtual memory in this situation.
3. Hardware Comparison: Create a comparison table for RAM, ROM, and Flash memory, considering their volatility and typical use cases.

4. Processor Architecture: Why is a multi-core processor generally more efficient at multitasking than a single-core processor running at a higher clock speed?
5. Technical Process: Outline the stages of the Fetch-Decode-Execute cycle.